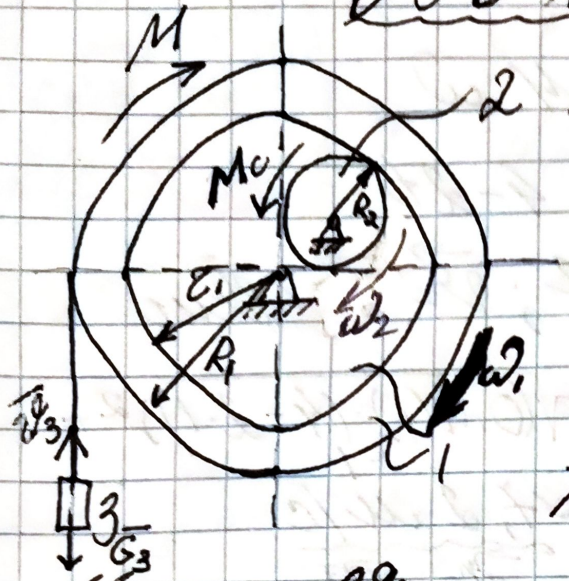
РГР №4 вар. 9



2 Дано: m_1, m_2, m_3;

$R_1, \varepsilon_1, i_i, R_2$,

$M = const$

$M_c = const$

$T_0 = 0, S.$

Найти: v_3

Решение: $T = \sum_{k=1}^{n} A_k^{ext}$

1) $T = T_1 + T_2 + T_3$; $T_1 = \frac{I_1 \omega_1^2}{2}$; $T_2 = \frac{I_2 \omega_2^2}{2}$; $T_3 = \frac{m_3 v_3^2}{2}$

$I_1 = \frac{m_1 R_1^2}{2}$; $I_2 = \frac{m_2 R_2^2}{2}$; $v_3 = \omega_1 R_1 \Rightarrow \omega_1 = \frac{v_3}{R_1}$ $\Rightarrow$

$\omega_2 R_2 = \omega_1 \varepsilon_1 \Rightarrow \omega_2 = \omega_1 \frac{\varepsilon_1}{R_2} = v_3 \frac{\varepsilon_1}{R_1 R_2}$

$\Rightarrow T = \frac{m_1 R_1^2}{2} \cdot \frac{v_3^2}{R_1^2} \cdot \frac{1}{2} + \frac{m_2 R_2^2}{2} \cdot \frac{v_3^2 \varepsilon_1^2}{R_1^2 R_2^2} \cdot \frac{1}{2} + \frac{m_3 v_3^2}{2} =$

$= \frac{v_3^2}{2}\left(\frac{m_1}{2} + \frac{m_2}{2} \cdot \frac{\varepsilon_1^2}{R_1^2} + m_3\right) = \frac{m_{пр} v_3^2}{2}$, $m_{пр}$

2) $\sum A_k^{ext} = A(\vec{G}_3) + A(M_c) + A(M) = A$

$A(\vec{G}_3) = -m_3 g S$ | $\frac{d\varphi_1}{dt} = \frac{dS}{dt} \cdot \frac{1}{R_1} \Rightarrow \varphi_1 = \frac{S}{R_1}$

$A(M_c) = -M_c \cdot \varphi_2$ | $\frac{d\varphi_2}{dt} = \frac{dS}{dt} \frac{\varepsilon_1}{R_1 R_2} \Rightarrow S \cdot \frac{\varepsilon_1}{R_1 \cdot R_2} = \varphi_2$

$A(M) = M \cdot \varphi_1$

$$A = -m_3 g S - M_c \cdot S \cdot \frac{r_1}{R_1 R_2} + M \frac{S}{R_1} =$$

$= S \cdot Q$, где $Q = \frac{M}{R_1} - M_c \cdot \frac{r_1}{R_1 R_2} - m_3 g$ – обобщённая сила

3) $\frac{m_{пр} v_3^2}{2} = Q \cdot S$

$$v_3 = \sqrt{\frac{2QS}{m_{пр}}}, \quad m_{пр} = \frac{m_1}{2} + \frac{m_2}{2} \frac{r_1^2}{R_1^2} + m_3$$

$$Q = \frac{M}{R_1} - M_c \cdot \frac{r_1}{R_1 R_2} - m_3 g$$